PS 2010: 7. Estimation and Inference 1

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Fall 2022

- Go to IPES conference if you can, **Next Friday and Saturday**
- Michelle Torres will visit us Nov 4th
- Meet with candidates
- Homework 5

Agenda

- Sampling and sampling distribution
- Bias and consistency
- Confidence interval

Random Sampling

- A **population** consists of the totality of the observations with which we are concerned.
- A **sample** is a subset of a population
- A **sample statistic** is a numerical measure describing some aspect of a sample
  - Sample mean and variance from repeated draws

  A **point estimator** is a rule (formula) for calculating a sample statistic that can be used as an estimate of a population parameter.
- The probability distribution of a statistic is called a **sampling distribution**

Sample Mean and Variance

Let x_1, x_2, x_3 ... x_n are randomly draw from a distribution, then

- Sample mean: $\bar{x} = \frac{1}{n}\sum_i^n x_i$
- Sample variance: $s^2 = \frac{1}{n-1}\sum_i^n (x_i - \bar{x})^2$
- Sample sd: $s = \sqrt{s^2}$

Example: A comparison of coffee prices at 4 randomly selected grocery stores in San Diego showed increases from the previous month of 12, 15, 17, and 20 cents for a 1-pound bag. Find the variance of this random sample of price increases.

Sampling Distribution

- $\bar{x}$ and s^2 depends on the underling distribution, the size of the sample, and the method of choosing the samples. Therefore, they are random variables
- The probability distribution of these random variables (we call them statistics) are sampling distribution
- Again, this sampling distribution comes from repeated sampling with the same size from the same population
- This sampling distribution help us to make inference about population parameters: population mean and variance

Sampling Distribution of Sample Mean

If Let X_1, X_2, X_3 ... X_n are randomly sample of size n from a normal distribution with mean μ and variance σ^2. Then

$$\bar{X} = \frac{1}{n}\sum_i^n X_i$$

is normally distributed with mean μ and variance $\frac{\sigma^2}{n}$

- We could write $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
- It also means we could transform it to $z = \frac{\bar{X}-\mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$

Central Limit Theorem

- In the above case, sampling distribution of $\bar{X}$ is normally distributed because we draw data from normal distribution
- However, this conclusion is generally true even we draw data from unknown distribution, provided that the sample size is large ($n \geq 30$)

Central Limit Theorem: If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution $N(0, 1)$.

Example

- Central Limit Theorem (CLT) has very important application in topics such as hypothesis testing, estimation, etc. Here are some examples:
- Achievement test scores of all high school seniors in a state have mean 60 and variance 64. A random sample of $n = 100$ students from one large high school had a mean score of 58. Is there evidence to suggest that this high school is inferior?
- An important manufacturing process produces a component parts for the automotive industry. An experiment is conducted in which 100 parts produced by the process are selected randomly and the diameter measured on each. It is known that the population standard deviation is 0.1 millimeter. What is the probability that the sample mean will be within 0.027 millimeter of the population mean.

Sampling Distribution of the Difference between Two Means

If independent samples of size n_1 and n_2 are drawn at random from two populations, discrete or continuous, with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2, respectively, then the sampling distribution of the differences of means, $\bar{X}_1 - \bar{X}_2$, is approximately normally distributed with mean and variance given by

$$\mu_{\bar{X}_1-\bar{X}_2} = \mu_1 - \mu_2 \text{ and } \sigma^2_{\bar{X}_1-\bar{X}_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Hence,

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{(\sigma_1^2/n_1) + (\sigma_2^2/n_2)}}$$

is approximately a standard normal variable.

Example

Given the following information on two population

Population A	Population B
$\mu_1 = 6.5$	$\mu_2 = 6.0$
$\sigma_1 = 0.9$	$\sigma_2 = 0.8$
$n_1 = 36$	$n_2 = 49$

What is the probability that a random sample of size 36 from A will have a mean that is at least 1 more than the mean of a sample size of 49 from B?

Sampling Distribution of Sample Variance

If S^2 is the variance of a random sample of size n taken from a normal population having the variance σ^2, then the statistic

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n} \frac{\left(X_i - \bar{X}\right)^2}{\sigma^2}$$

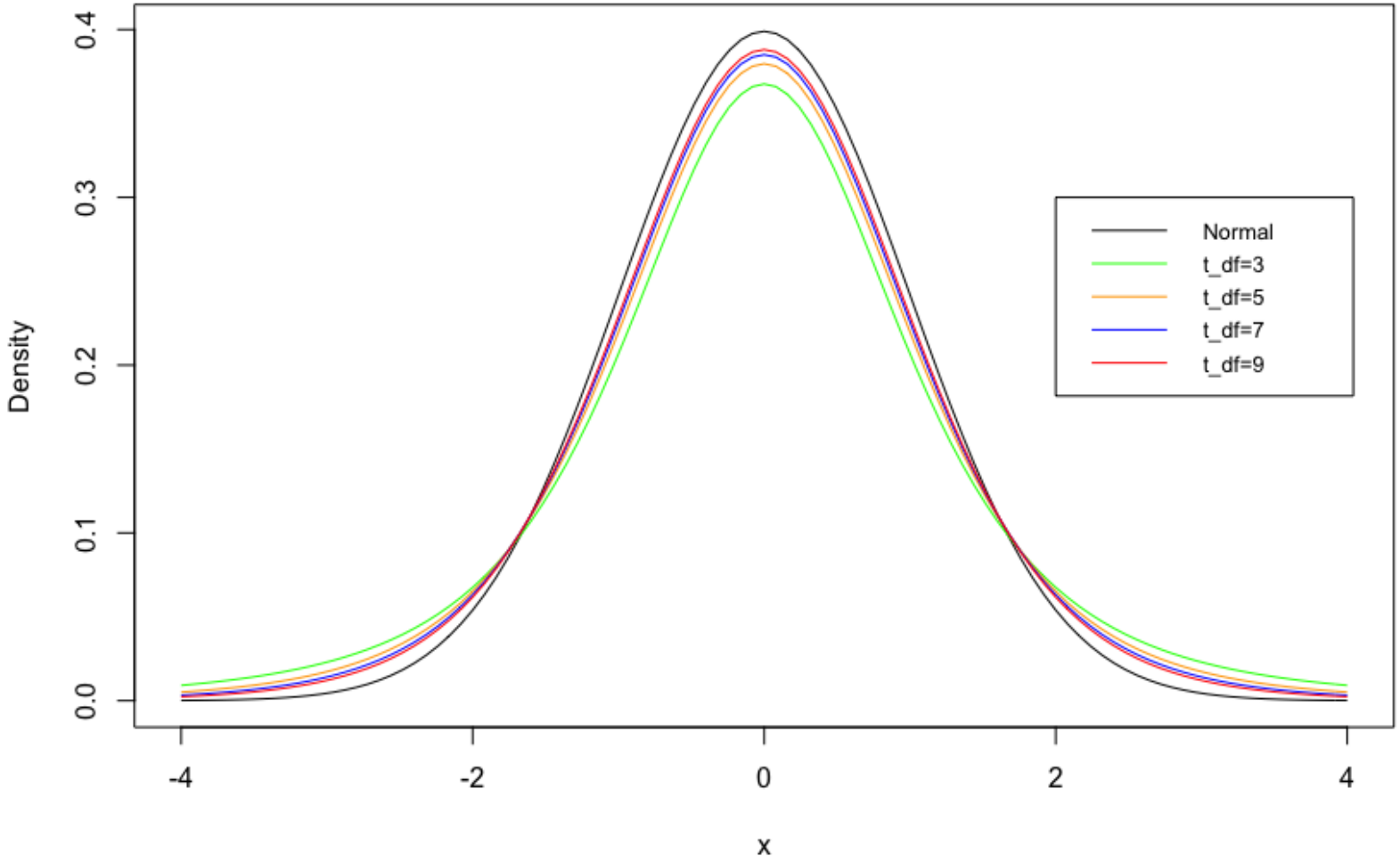
has a chi-squared distribution with $n-1$ degrees of freedom.

- Notice: $z - score = \frac{X-\mu}{\sigma}$

Student t Distribution Revisit

- In the sampling distribution of sample mean, we assume we know σ
- However, this is not the case for many applications
- We could use the sample standard deviation s to substitute σ
- If the population distribution is normal, this makes the sampling distribution of the sample mean as a student t distribution
- When sample size ≤ 30, better to use student t
- When sample size > 30, ok to use standard normal

t distribution



Example

A chemical engineer claims that the population mean yield of a certain batch process is 500 grams per milliliter of raw material. To check this claim he samples 25 batches each month. If the computed t-value falls between $-t_{0.05}$ and $t_{0.05}$, he is satisfied with this claim. What conclusion should he draw from a sample that has a mean $\bar{x} = 518$ grams per milliliter and a sample standard deviation $s = 40$ grams? Assume the distribution of yields to be approximately normal.

Statistical Inference

- **Statistical inference** involves drawing conclusions about features of an underlying population using relatively small samples of data from that population.
  - Frequentist
  - Bayesian
- Estimation and Hypotheses testing
- A **point estimate** of a population parameter is a single value of a statistic
  - Examples: sample mean
- However, an estimator is not expected to estimate the population parameter without error.
  - Sample size, sample value, sample method

Bias

- To measure a goodness of an estimator we can use bias measure
- An unbiased point estimator produces a sample statistic with sampling distribution that has a mean equal to the population parameter the statistic is intended to estimate.

  Let $\hat{\theta}$ be a point estimator for a parameter θ. Then $\hat{\theta}$ is an unbiased estimator if $E(\hat{\theta}) = \theta$. If $E(\hat{\theta}) \neq \theta, \hat{\theta}$ is said to be biased.
- The bias of a point estimator $\hat{\theta}$ is given by $B(\hat{\theta}) = E(\hat{\theta}) - \theta$

Example

If X_i are normally distributed random variables with mean μ and variance σ^2, then:

$$\bar{X} = \frac{\sum X_i}{n} \text{ and } S^2 = \frac{\sum \left(X_i - \bar{X}\right)^2}{n}$$

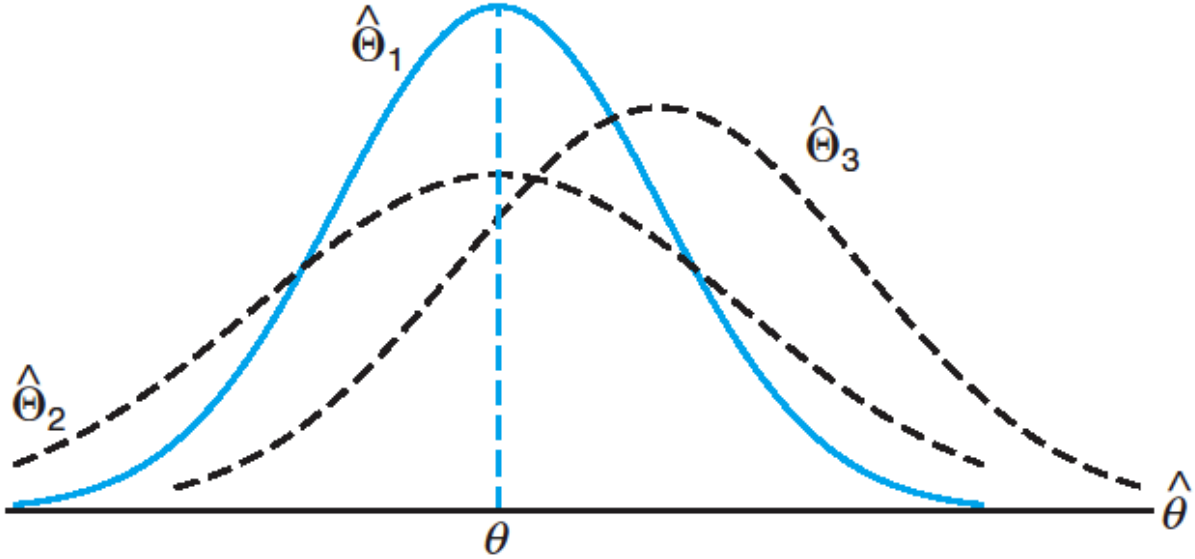
- Prove that $\bar{X}$ and S^2 are biased/unbiased estimator for μ and σ^2
- If it is biased estimator, what is the bias?

Mean Square Error

- Another way to measure the goodness of an estimator in mean square error (MSE). The MSE for $\hat{\theta}$ is

$$\text{MSE}(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right].$$

$$\begin{aligned}
\text{MSE} &= \mathbb{E}\left\{(\hat{\theta} - \theta)^2\right\} \\
&= \mathbb{E}\left[\{(\hat{\theta} - \mathbb{E}(\hat{\theta})) + (\mathbb{E}(\hat{\theta}) - \theta)\}^2\right] \\
&= \mathbb{E}\left[\{\hat{\theta} - \mathbb{E}(\hat{\theta})\}^2\right] + \{\mathbb{E}(\hat{\theta}) - \theta\}^2 \\
&= \text{ variance } + \text{ bias }^2.
\end{aligned}$$


$\hat{\Theta}_1$
$\hat{\Theta}_3$
$\hat{\Theta}_2$
θ
$\hat{\theta}$

Property of Point Estimator: Efficiency and Consistency

- MSE cares about both variance and bias
- **Efficiency**, we want an estimator is unbias and has the smallest variance among all unbiased estimator.
- RMSE is just the root of MSE
- A consistent point estimator converges in probability to the true parameter value.

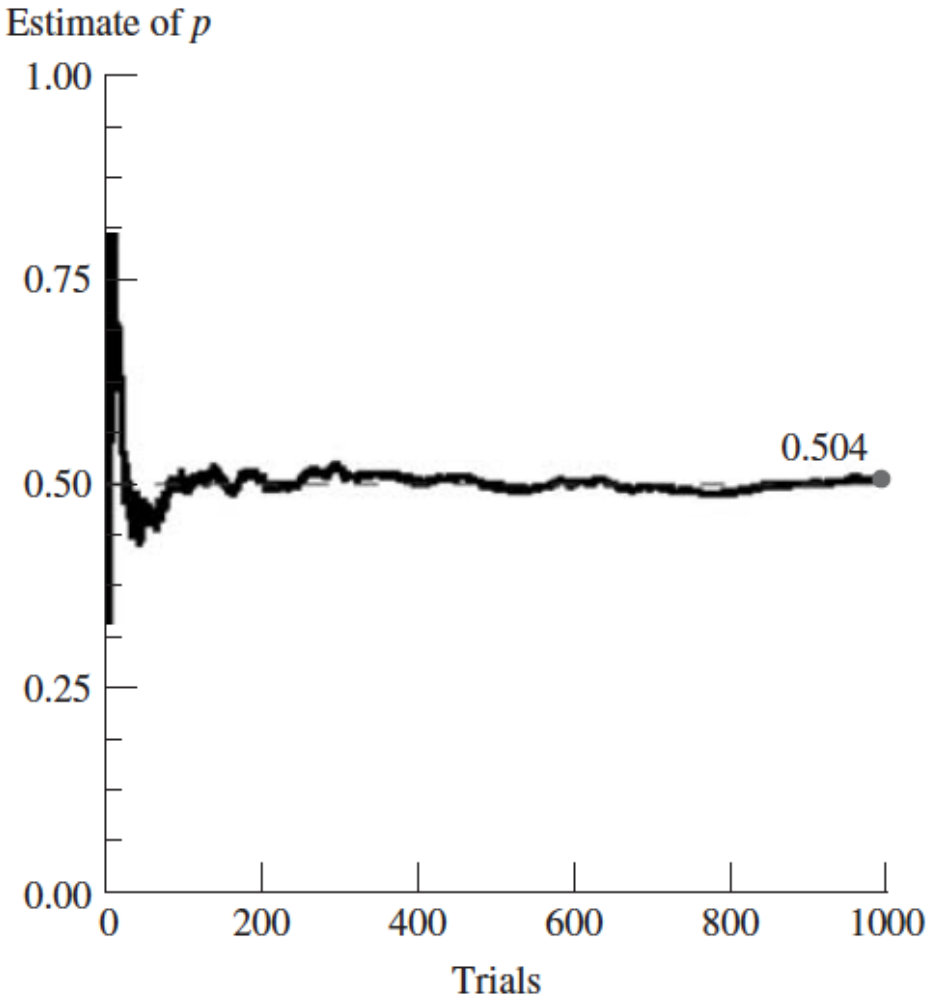
$$\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta\right| \leq \varepsilon\right) = 1$$

or, equivalently,

$$\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta\right| > \varepsilon\right) = 0$$

Understanding Consistency

- Flip a coin 5, 10, 20, 100, 1000 times and calculate $P(Head)$



Unbiasedness and Consistency

An unbiased estimator $\hat{\theta}_n$ for θ is a consistent estimator of θ if

$$\lim_{n \to \infty} V\left(\hat{\theta}_n\right) = 0$$

- Example: Let $Y_1, Y_2, \ldots, Y_n$ denote a random sample from a distribution with mean μ and variance $\sigma^2 < \infty$. Show that $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$ is a consistent estimator of μ.

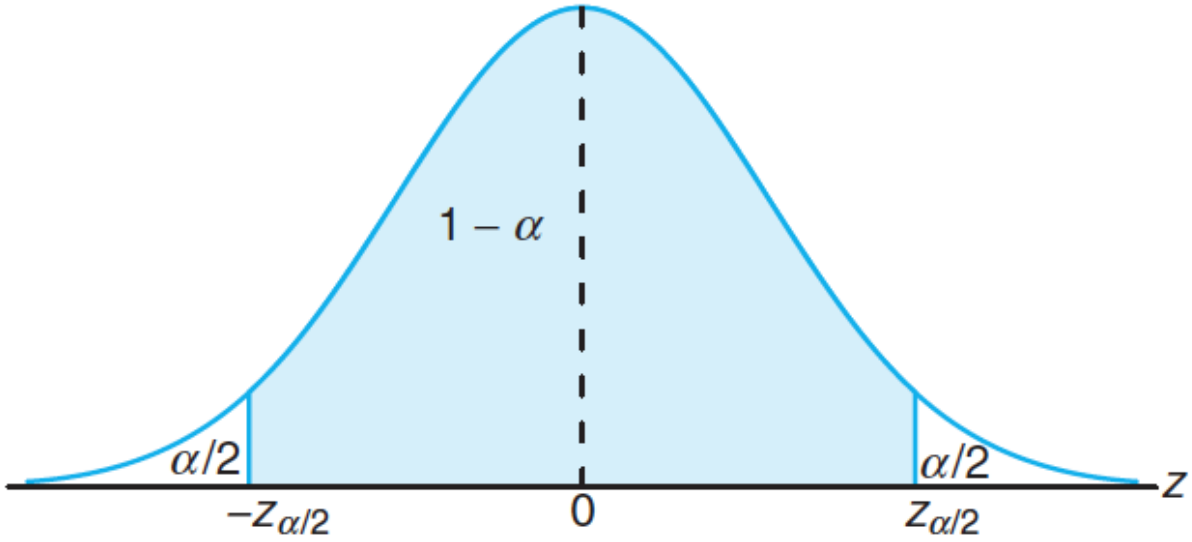
Confidence Interval

- We have learned how to measure a goodness and properties of an estimator
- Another way of measuring uncertainty of an estimator is confidence interval
- It involves of characterizing and describing the sampling distribution of statistics
- Confidence intervals give a range of values that are likely to include the true value of the parameter.
  - over a hypothetically repeated data-generating process
  - contain the true value of the parameter
  - with the probability specified by the confidence level

Confidence Interval

- Also called confidence bounds, error bands, interval estimators
- Two sided CI have lower and upper bounds. One sided CI is also allowed
- Research need to decide the confidence level by $1 - \alpha$

$$P\left(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U\right) = 1 - \alpha,$$



Constructing CI for Large Sample

- Under CLT, the sampling distribution of an estimator is normally distributed
- We could then transform it to standard normal
- That is $Z = \frac{\hat{\theta}-\theta}{\sigma_{\hat{\theta}}}$
- If we want CI as $P\left(-z_{\alpha/2} < Z < z_{\alpha/2}\right) = 1 - \alpha$
- Then, $P\left(-z_{\alpha/2} < \frac{\hat{\theta}-\theta}{\sigma_{\hat{\theta}}} < z_{\alpha/2}\right) = 1 - \alpha$
- Finally

$$P\left(\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}} < \theta < \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}}\right) = 1 - \alpha$$

- The CI is

$$\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}} < \theta < \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}}$$

CI for μ in Large Sample

If $\bar{x}$ is the mean of a random sample of size n from a population with known variance σ^2, a $100(1-\alpha)\%$ confidence interval for μ is given by

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right.

- If σ is unknown, use s sample standard deviation to replace it.
- However, this is not ok under small sample size

Examples

- The above are general procedures, let's look at some examples
- SAT mathematics scores of a random sample of 500 high school seniors in the state of Texas are collected, and the sample mean and standard deviation are found to be 501 and 112, respectively. Find a 99% confidence interval on the mean SAT mathematics score for seniors in the state of Texas.
- The shopping times of n $= 64$ randomly selected customers at a local supermarket were recorded. The average and variance of the 64 shopping times were 33 minutes and 256, respectively. Estimate μ, the true average shopping time per customer, with a confidence level of $1 - \alpha = .90$.

Summary

- Under large sample, we have no assumption on the population distribution
- CLT makes sure sampling distribution of estimators have normal distribution
- We can also safely replace σ using s and using standard normal for the critical value
- However, using s to replace σ is student t in small sample size if sample comes from normal distribution

Constructing CI for Small Sample

- Notice, in the small sample and the variance of population is unknown, the sampling distribution is not normally distributed any more
- Under the assumption that sample is randomly draw from normal distribution
- We need to use student t distribution to construct the CI for population mean μ
- The rest of the procedures are the same

CI for μ in Small Sample

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal population with unknown variance σ^2, a $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{x} - t_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2}\frac{s}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the t-value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

Example

A meat inspector has randomly selected 30 packs of 95% lean beef. The sample resulted in a mean of 96.2% with a sample standard deviation of 0.8%. Find a 99% prediction interval for the leanness of a new pack. Assume normality.

CI for Proportion

- In a binomial experiment, the proportion of success can be represented by $\hat{p} = \frac{x}{n}$
- For large sample, under CLT, the variance of sampling distribution is $\frac{\sigma^2}{n} = \frac{n\hat{p}\hat{q}}{n}$
- Therefore, we could have CI for proportion as

$$P\left(\widehat{P} - z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \widehat{P} + z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}\right) \approx 1 - \alpha$$

- This is large sample result

Example

In a random sample of $n = 500$ families owning television sets in the city of Hamilton, Canada, it is found that $x = 340$ subscribe to HBO. Find a 95% confidence interval for the actual proportion of families with television sets in this city that subscribe to HBO.